

Project Proposal White Paper

ME366 :Electro-Mechanical System Design of a Low-Cost Smart Astronomical Telescope

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Abstract

This white paper proposes the design and localized manufacturing of a low-cost, smart astronomical GoTo telescope using alternative materials, optomechanical structural optimization, and indigenous fabrication techniques.

1 Summary

Because the current availability of astronomical telescopes in Bangladesh relies entirely on expensive imports, this project aims to democratize the niche hobby of celestial observation for the local population through domestic, cost-effective manufacturing.

2 Project Category (in compliance with ME-366)

Robot Autonomy, Measurement, Control, and Electromechanical Test Rigs.

3 Architecture & Technical Description

- **Kinematic Autonomy:** Controlled by a microcontroller, the telescope will feature two degrees of freedom (Alt-Azimuth coordinates). Upon receiving commands from a smartphone interface, it will automatically point at target celestial bodies. Additionally, it will counteract the Earth's diurnal rotation via constant, micro-stepping tracking driven by high-precision stepper motors and a custom gear-train assembly.
- **Optical Configuration:** Refractor type, which will utilize an achromatic doublet lens system as the primary objective to minimize chromatic aberration.
- **Imaging and Viewing:** Prime focus imaging coupled with live-view streaming to a smartphone over a local Wi-Fi network.
- **Structural Materials:** The chassis and components will be fabricated mostly using 3D-printed polymers and rigid PVC materials.

4 Mapping with Complex Engineering Problem (CEP) Criteria

4.1 P-1: Depth of Knowledge

The project will demand integrated knowledge across advanced geometric optics, precision structural mechanics for stable tracking and pointing, and electromechanical system design for hardware-software synchronization.

4.2 P-2: Conflicting Requirements

- Rigidity vs. Weight.
- Size vs. Light Gathering Ability.

4.3 P-3: Depth of Analysis

Capturing faint starlight from deep-sky objects and resolving it onto a digital sensor for real-time mobile transmission will demand rigorous analytical effort.

4.4 P-7: Interdependence of Subsystems

The system execution will flow from Target designation → real-time electromechanical pointing and sidereal tracking → continuous digital image capture and streaming.

5 Approximate Cost Analysis

Component	Description	Est. Cost (BDT)
Optical Elements	Achromatic doublet lens	7000
Actuators & Drive	Stepper motors, custom 3D-printed gears, bearings	2500
Electronics & IoT	Microcontroller, motor drivers, Wi-Fi camera module	4000
Structure & Chassis	PVC tubes, 3D printing filament, fasteners, tripod	3500
Miscellaneous		2000
Total Estimated Cost		19000 BDT